A Decision Support System for Agricultural Profit-Sharing: Design, Risk Analysis, and Performance Evaluation

UNDERGRADUATE THESIS

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INFORMATICS STUDY PROGRAM
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HORIZON INDONESIA UNIVERSITY
2027

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We, the undersigned, hereby declare that the undergraduate thesis entitled:

A Decision Support System for Agricultural Profit-Sharing: Design, Risk Analysis, and Performance Evaluation

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If it is later proven that this thesis contains elements of plagiarism, we are willing to accept that this thesis be declared invalid, that any academic degree obtained be revoked, and that we be processed in accordance with applicable laws and regulations.

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Hani Ayu Fadila	Liana Syifa Fauzia	Ririn Dwi Ariyanti

APPROVAL PAGE

THESIS TITLE

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Nama Penguji 2
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ABSTRACT

The Indonesian agricultural sector faces major challenges, especially from the side of farmers and investors. Where farmers who own land do not have the capital to manage their land optimally, while investors who have capital are hesitant to invest with farmers, because of uncertainty regarding the expected results will not be in accordance with what is expected. Many cases where investors actually suffer large losses from investments, so a digital tool is needed that can help the risk of crop failure for investors and help farmers get capital to manage their land, without having to accept losses from both parties. This study aims to design and develop a mobile and web-based agricultural platform using the Design Thinking method, which is equipped with a Decision Support System feature for profit sharing with an AI Risk Estimator as its main feature. The system uses a rule-based artificial intelligence approach to predict the risk of crop failure based on data on crop types, land location, agricultural history, and weather conditions. This platform is expected to help investor farmers transparently through a contract system for digital profit sharing with real-time performance tracking. System testing was carried out using Black Box Testing and System Usability Scale (SUS) to evaluate functionality and ease of use. This research is expected to provide real solutions for farmers' access to capital while providing a safer and more transparent agricultural investment experience for investors.

Keyword: Decision Support System, Profit-Sharing, AI Risk Estimator, Agricultural Platform

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Acknowledgments

In the **Acknowledgments** section, thank individuals who contributed to the success of the project. The **number of acknowledgments page** is adjusted to the **number of thesis authors**. This could include:

1. Your advisor(s), fellow researchers, peers, or anyone who helped in the research, development, or writing process.
2. A brief description of the contributions made by these individuals.

PREFACE

First and foremost, praise and gratitude are extended to Allah SWT for His blessings and grace, which have enabled the author to complete this undergraduate thesis report. This report is submitted as one of the requirements for obtaining a Bachelor's degree (S1) in the Informatics Study Program, Faculty of Information Technology and Computer Science, Horizon Indonesia University.

The author realizes that this thesis report is still far from perfect. Therefore, constructive criticism and suggestions are highly appreciated. Despite all limitations, the author also acknowledges that this thesis report would not have been possible without the support, guidance, and encouragement from various parties. Thus, with sincere gratitude, the author would like to express appreciation to:

1. …… as the Rector of Horizon Indonesia University;
2. …… as the Dean of the Faculty of Information Technology and Computer Science:
3. …… as the Head of the Study Program …… *(sesuaikan dengan program studi penulis)*
4. …… as the First Thesis Advisor;
5. …… as the Second Thesis Advisor;
6. The author’s beloved parents *(bisa disesuaikan penulisannya)*
7. …… *(selanjutnya dapat disesuaikan dengan kebutuhan penulis)*

May Allah SWT reward all kindness and always bestow His guidance, blessings, and grace upon them. Aamiin.

Karawang, January 2, 2025
(disesuaikan dengan tanggal dikumpulnya laporan skripsi)

(tanda tangan penulis)

Anwar Hilman
(nama penulis)

Table of Contents

The **Table of Contents** lists all the chapters, sections, and subsections with page numbers. The format should look as follows:

1. **Font**: 12 pt, Times New Roman
2. **Spacing**: 1.5-line spacing
3. **Alignment**: Left-aligned, but the page numbers should be right-aligned.

The Table of Contents should auto-generate if using a word processor with heading styles (e.g., Microsoft Word).

List of Figures and Tables

If your thesis includes figures and tables, include a list after the table of contents. For each item, provide the figure/table number and title.

Example:

List of Figures

1. Figure 1: System Architecture Design
2. Figure 2: User Interface Prototype

List of Tables

1. Table 1: Performance Test Results
2. Table 2: Database Schema Overview

The List of Figures and Tables should auto-generate

Chapter 1: Introduction

1.1 Background

The agricultural sector is widely recognized as one of the most important sectors in Indonesia's economy. However, despite its importance, it still faces a number of ongoing structural problems. One issue that often occurs is the gap between farmers and investors. Many farmers already have access to land, but they do not have enough capital to manage it properly. Meanwhile, investors who actually have the financial resources tend to be more careful when it comes to investing in agriculture. This mainly happens because agriculture involves a high level of uncertainty, particularly due to the risk of crop failure.

As a result, much of the available agricultural land is not utilized to its full potential. In addition, many investment opportunities that could have been beneficial are not taken advantage of. In the long run, this condition affects not only agricultural output but also the economic development of rural areas. There have been several digital solutions developed to respond to this issue. However, most of them still focus on certain parts of the problem and have not completely addressed the gap between farmers and investors.SIPROTANI (Dewi et al., 2022) developed a mobile-based agricultural information system focused on documenting and reporting agricultural products using the Agile/Scrum method, yet it does not address financial transactions, investment mechanisms, or risk assessment between farmers and investors. Similarly, the Petani Cerdas application (Wijaya et al., 2024) provides agricultural training and consultation features for farmers using an evolutionary prototype method, but lacks any investment system or AI-based risk estimation capability. A broader study by Prayoga (UGM) reviewing various agricultural digital platforms in Indonesia, including iGrow and TaniHub, found that while some platforms offer profit-sharing investment concepts, none integrate an automated AI risk estimator accessible directly by investors. Furthermore, research by Palupi et al. (2021) analyzing the fintech platform Crowde revealed that crop failure risk, referred to as Non-Performing Loan (NPL), remains a critical challenge, yet risk assessment is still conducted manually by the company's operational team rather than through an automated digital system. Most recently, Khuluq and Sriyono (2025) emphasized the strategic importance of integrating financial technology with digital agricultural platforms, noting that most existing studies address each domain separately without providing an integrated system featuring intelligent risk assessment.

This research therefore aims to design and develop a mobile and web-based agricultural platform using the Design Thinking methodology, integrating a Decision Support System with an AI Risk Estimator as its core feature. The system utilizes a rule-based artificial intelligence approach to predict crop failure risks based on plant type, land location, agricultural history, and weather conditions, while also facilitating a transparent digital profit-sharing contract system between farmers and investors with real-time performance tracking. This solution is expected to bridge the existing gap in agricultural technology by providing a safer, more transparent, and data-driven investment ecosystem that empowers both farmers and investors in Indonesia.

1.2 Problem Statement

Based on the background described above, the research problem can be defined as follows: How can a mobile and web-based agricultural platform be developed using the Design Thinking methodology that integrates a Decision Support System with an AI Risk Estimator to predict crop failure risks and support a transparent digital profit-sharing system between farmers and investors?

1.3 Research Objectives

Based on the problem identification previously established, the objectives of this research are:

1. To design and develop a mobile and web-based agricultural platform using the Design Thinking methodology that integrates a transparent digital profit-sharing system between farmers and investors.
2. To integrate a Decision Support System (DSS) featuring a rule-based AI Risk Estimator capable of predicting crop failure risks based on crop type, land location, agricultural history, and weather conditions.
3. To construct a real-time agricultural performance tracking system to provide accurate data for investors in making informed investment decisions.
4. To validate the effectiveness of an intelligent system in mitigating investment risks (reducing Non-Performing Loans) while optimizing the utilization of agricultural land in Indonesia.

1.4 Research Benefits

This study is built to close the gap between farmers and investors through smart technology. We've broken down the benefits into two categories:

1. Academic Contributions
This research adds fresh perspectives to Agricultural IT and Decision Support Systems by:

a. Offering a transparent scholarly reference regarding how the Design Thinking model can address complex issues involving multiple participants in agriculture.
b. Assisting in the development of more intelligent agricultural failure predictions by employing artificial intelligence to evaluate past farming practices, data unique to certain locations, and climatic patterns.
c. Delivering a robust scientific method for evaluating digital platforms, guaranteeing their seamless functionality and delivering an excellent experience for the user.

2. Practical Contributions
On the ground, this research provides tools that make a real difference in the digital farming world:

a. Farmers

It paves the way for smoother financial support by eliminating excessive bureaucracy. With enhanced availability of funds, agriculturalists can concentrate on optimizing their land and utilizing superior resources to boost their production.

b. Investors
The AI Risk Estimator removes uncertainty from the investment process. It provides investors with clear, data-based information prior to making any financial commitments, assisting them in avoiding poor transactions and reducing possible losses.

c. Farming Ecosystem
With real-time performance tracking, the entire process becomes transparent. This builds genuine trust between investors and farmers, cuts out communication gaps, and makes the whole food supply chain run much more effectively.

1.5 Problem Scope

This study focuses on the development of a Decision Support System (DSS) for profit-sharing schemes in the agricultural sector, specifically to assist investors in assessing investment risks and to help farmers communicate information regarding agricultural conditions.

This research addresses several key issues, including:

1. a digital system connecting farmers and investors to enhance information transparency between them within the profit-sharing scheme
2. an AI Risk Assessment feature to estimate the potential for crop failure, which currently makes it difficult for investors to assess this risk before investing
3. An information dashboard to support investment decisions, as there is currently no system capable of providing risk estimates as a basis for decision-making

The limitations not covered in this research include:

1. Does not address system implementation at a production scale or real-world field applications
2. Does not cover direct financial transactions
3. Does not address the physical distribution of harvests or logistics
4. Does not use complex machine learning-based AI models (only rule-based)
5. Additional features such as marketplaces and donations are not the primary focus of this research

The prototype to be developed in this research consists of:

1. A web application used by investors to:
    a. View agricultural project information
    b. View risk analysis results
    c. Support the decision-making process

2. A mobile application used by farmers to:

    a. Input agricultural data
    b. Update the condition and progress of crop fields

1.6 Research Methodology

This study employs the design thinking method, an approach that focuses on user needs when designing solutions to a problem. The stages of the Design Thinking method used are as follows:

1. **Empathize**

   This stage aims to understand the needs and challenges of the users, namely farmers and investors. Activities conducted in this stage include a literature review on agricultural profit-sharing systems and an analysis of issues such as the risk of crop failure and information gaps.

2. **Define**

   In this stage, the problem is formulated based on the results of the Empathize stage. The primary issues identified are a lack of transparency in information between farmers and investors, the absence of a system to assist investors in assessing investment risks, and the lack of a decision-support system within the agricultural profit-sharing scheme.

3. **Ideate**

   This stage involves developing solution ideas to address the problems defined in the Define stage. Proposed solutions include the development of a digital system to connect farmers and investors, the integration of AI-based risk assessment features for estimating crop failure risks, and the provision of a dashboard as a decision-making tool.

4. **Prototype**

   In this phase, the design and development of the system prototype are carried out. The developed prototype includes a web application for investors and a Flutter-based mobile application for farmers, with the prototype development covering user interface (UI/UX) design, system flow design, and the implementation of key features such as the profit-sharing system and risk analysis.

5. **Test**

   This phase is conducted to test the developed system to determine whether it operates in accordance with the research objectives, and the results of this phase are used as the basis for evaluating and improving the developed system. The testing methods used

include Black Box Testing to assess system functionality and the System Usability Scale (SUS) to measure usability.

1.7 Report Structure

Provide a roadmap of the thesis structure, summarizing each chapter briefly.

Chapter 2: Literature Review

2.1 Key Definitions and Concepts

Define key technical terms related to your project (e.g., cloud computing, network security, or programming paradigms) to ensure clarity.

2.2 Related Work

Summarize relevant studies, research papers, and projects that are related to your work. Highlight how they influenced your approach.

2.3 Technologies Used

Provide a deep dive into the technologies used in your project, including frameworks, tools, or libraries. Explain why each was chosen based on the project's goals.

2.4 Theoretical Framework

Explain the theory, models, or systems that your work is based on. This could include technical frameworks, algorithms, or methodologies.

Chapter 3: System Design and Implementation

3.1 System Design

Describe the overall system design. Include diagrams (e.g., UML, data flow diagrams) and explain how the system components interact.

3.2 System Architecture and Components

Detail the architecture of the system (e.g., microservices, monolithic). Discuss each component's function within the system and its technical design.

3.3 Technologies and Tools Used

Provide an explanation of the software, hardware, or tools used for system development and testing. Mention the rationale behind choosing each technology.

3.4 System Development and Testing

Explain how the system was built, tested, and debugged. Include any frameworks, programming languages, or methodologies used during development.

Chapter 4: Results and Discussion

4.1 System Implementation Results

Present the outcomes of the project, including features developed and system functionality.

4.2 Discussion and Analysis of Results

Analyze the system's performance. Was it able to solve the technical problems as expected? Include any lessons learned during implementation.

4.3 System Performance Evaluation

Evaluate system performance metrics such as speed, scalability, security, and usability. Use quantitative data if possible (e.g., performance benchmarks).

Display in table form

Example:

Table 1. System Performance

No	Aspect	Old System	New System
1	Data Redundancy	Many redundant data	Zero redundant data
2	Monthly Report Generation	3 days	< 2 minutes

4.4 Comparison with Related Projects

Compare your project with similar work in the field. What makes your solution unique? How does it perform relative to other systems?

Display in table form

Example:

Table 2. Comparison Project

No	Comparison Aspect	Another Project	This Project
1	Data collection time span	1 month	2 month
2	Type of Programming	Structured	Object Oriented

Chapter 5: Conclusions and Recommendations

5.1 Conclusions

1. Summarize the key findings of the project and how it met the research objectives. Discuss the system's final impact and usefulness.
2. Ensure **Conclusions align with Research Objectives**

5.2 Recommendations

1. Provide suggestions for future research or improvements to the project. This could involve scaling the system, adding features, or using new technologies.
2. Ensure **Recommendations align with Problem Scope** (Research Limitations)

References

List all sources referred to in your thesis, following a consistent citation style (e.g., **IEEE** or **APA**). Include research papers, books, online resources, and any other references used in your work.

1. Guidance Reference:

a. Students are required to cite relevant articles from lecturers, whether books, journals, proceedings or online writings, at least 2 articles from supervisors and 3 articles from faculty lecturers.
b. For writing references, students are required to take sources from Books, Journals, Proceedings, at least 35 sources. The other 3 international sources may be from national sources.
c. Regulations regarding sources for writing quotations from other sources use IEEE standards and use Mendeley and Zotero Reference Management Software.

2. **Example references guidance**:

A. Buku

1) Satu Penulis :

Format :

[Nomor Referensi] Penulis, Judul Buku Edisi, Tempat Penerbitan, Penerbit, Tahun.

Contoh :

[1] A. V. Oppenheim and R. W. Schafer, Pemrosesan Sinyal Waktu Diskrit Edisi ke-3. Upper Saddle River, NJ: Prentice Hall, 2010.

2) Dua Penulis :

Format :

[Nomor Referensi] Inisial Penulis 1. Nama Belakang dan Inisial Penulis 2. Nama belakang, Judul Buku Edisi, Tempat Penerbitan, Penerbit, Tahun.

Contoh :

[1] A. H. Smith dan J. K. Johnson, Pemrosesan Sinyal Digital Edisi ke-4. New York, NY: McGraw-Hill, 2019.

3) Lebih dari 2 Penulis :

Format :

[Nomor Referensi] Inisial Penulis 1. Nama Belakang, Inisial Penulis 2. Nama Belakang, Inisial Penulis 3. Nama Belakang, dkk., Judul Buku Edisi, Tempat Penerbitan, Penerbit, Tahun.

Contoh :

[1] A. B. Johnson, L. R. Smith, S. M. Patel, et al, Komunikasi Digital Edisi ke-5. New York, NY: McGraw-Hill, 2022.

B. e-Book

Format :

[Nomor Referensi] Penulis, Judul Buku Nomor (jika ada), Tempat Publikasi: Penerbit, Tahun Penerbitan. [Online]. Tersedia: URL atau DOI.

Contoh :

[1] G. J. Myers, Seni Pengujian Perangkat Lunak Edisi ke-3. New York: John Wiley & Sons, Inc, 2012. [Online]. Available: https://doi.org/10.1002/9781118411554

C. Journal article

Format :

[Nomor Referensi] Penulis, "Judul Artikel," Judul Jurnal yang Disingkat Nomor, vol. nomor, no. terbitan, hal. nomor halaman, Bulan dan tahun.

Contoh :

[1] R. L. Aggarwal dan V. N. Tiwari, "Protokol perutean hemat energi untuk jaringan sensor nirkabel," Int. J. Comput. Sci. Eng., vol. 4, no. 1, hal. 97-104, Januari 2012.

D. Artikel e-Jurnal

Format :

[Nomor Referensi] Penulis, "Judul Artikel," Judul Jurnal vol. nomor volume, no. nomor terbitan, hal. nomor halaman, Bulan Tahun. Tersedia: DOI atau URL

Contoh :

[1] J. Smith dan M. Johnson, "Penggunaan kecerdasan buatan dalam perawatan kesehatan," Jurnal Penelitian Internet Medis, vol. 22, no. 3, hal. e16260, Maret 2020. Tersedia: https://doi.org/10.2196/16260.

E. Prosiding

Format :

[Nomor Referensi] Penulis, "Judul Makalah," dalam Nama Konferensi, Tempat Konferensi, Tahun, hlm. nomor halaman. DOI atau URL

Contoh :

[1] J. Doe dan A. Smith, "Pendekatan baru untuk pengenalan gambar menggunakan pembelajaran mendalam," dalam Prosiding 2019 Konferensi Internasional tentang Pembelajaran Mesin New York, NY, USA, 2019, hlm. 112-119. doi: 10.1109/ICML.2019.00016.

F. Tesis atau disertasi

Format :

[Nomor referensi] Inisial penulis. Nama belakang penulis, "Judul tesis," Jenis sebutan, Abbreviation. Departemen, Singkatan, Abbrev. Universitas, Kota Universitas, Negara Bagian, Tahun.

Contoh :

[1] J. Smith, "Investigasi Efisiensi Panel Surya di Lingkungan Ekstrem," Disertasi Ph.D., Departemen Teknik Elektro, Universitas California, Los Angeles, CA, USA, 2021.

G. Situs web

Format :

[Nomor Referensi] Penulis, "Judul Halaman Web," Judul Situs Web, [Online]. Tersedia: URL. [Diakses: Tanggal].

Contoh :

[1] A. Smith, "Sejarah Fotografi," PhotographyLife, [Online]. Tersedia: https://photographylife.com/history-of-photography. [Diakses: 15 Maret 2023].

Appendices

1. **Research Documentation**

   Include supporting materials such as source code, diagrams, system configurations, testing results, or any other relevant data.

2. **Researcher Curriculum Vitae**

   The curriculum vitae page contains the author's name, place and date of birth, parents' names, educational history and work history as well as prominent achievements. Presented in a creative form but does not give up formal elements. The photos used also use formal photos, not selfie photos.

3. **Research Letter**

   A Research letter from Horizon University Indonesia must be included. If the research object is an agency/institution/other official institution, then a reply letter/research statement must be included. If the research object is not an agency/institution/institution, then the reply letter/information is not required to be included.